# Week 2 Penetration Testing Report: Footprinting & Network Scanning

**Target:** networkwalks.com & Local Subnet

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**Batch:** B084
**Program:** Networkwalks Cybersecurity & Ethical Hacking Internship
**Instructor:** Waqas Karim CCIE
**Date:** 08/11/2026

**Repository:**https://github.com/Mastertactician23/NETWORKWALKS-KOFFI-B084-WK2-FINAL-REPORT

## 1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

## 2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1) and scanning my own local network with Zenmap (W2-PM5). One module covers the passive footprinting phase and the other covers the active scanning phase, demonstrating how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 final report of my ongoing internship program at Networkwalks.

All commands were run in Kali Linux (footprinting) and on a Windows PC with Zenmap installed (scanning). Every step below includes the exact command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

## 3. Tools Used

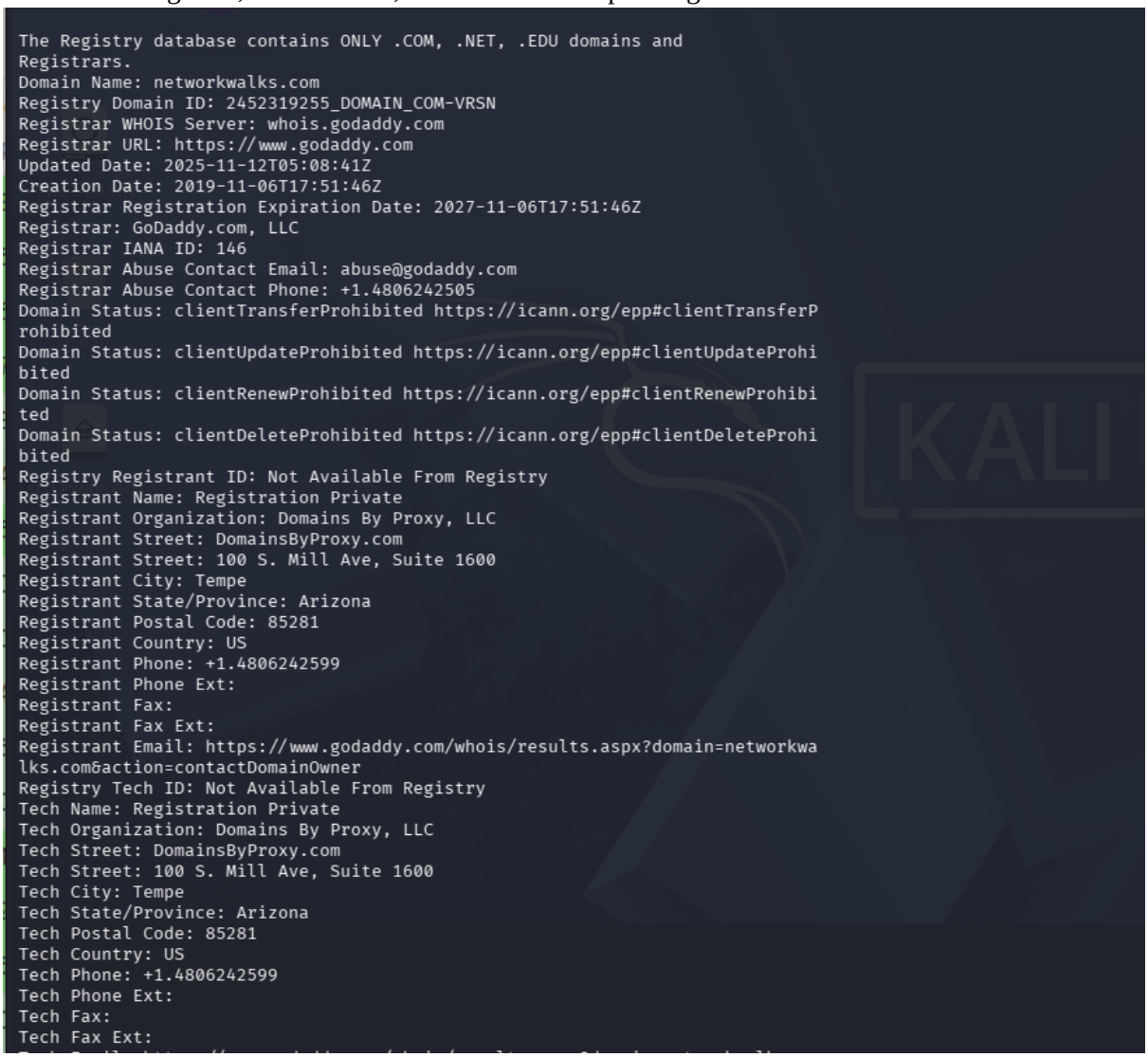
| Tool | Purpose |
|---|---|
| Kali Linux & Windows | Operating systems used for reconnaissance activities |
| WHOIS | Find domain registration details (owner, dates, name servers). |
| whatweb | Fingerprint web technologies (server, CMS, plugins, IP). |
| nslookup | Resolve the domain name to its IP address using DNS. |
| curl -I | Read the HTTP response headers of the website. |
| wafw00f | Detect whether a Web Application Firewall protects the site. |
| dnsrecon | Enumerate all DNS records (NS, MX, SPF, TXT, SRV). |
| Zenmap (Nmap GUI) | Scan the local subnet to find live hosts, IPs and MAC addresses. |
| Windows CMD | Local IP and MAC address identification |

## 4. Activities Performed

### 4.1 Footprinting & Reconnaissance (W2-PM1)

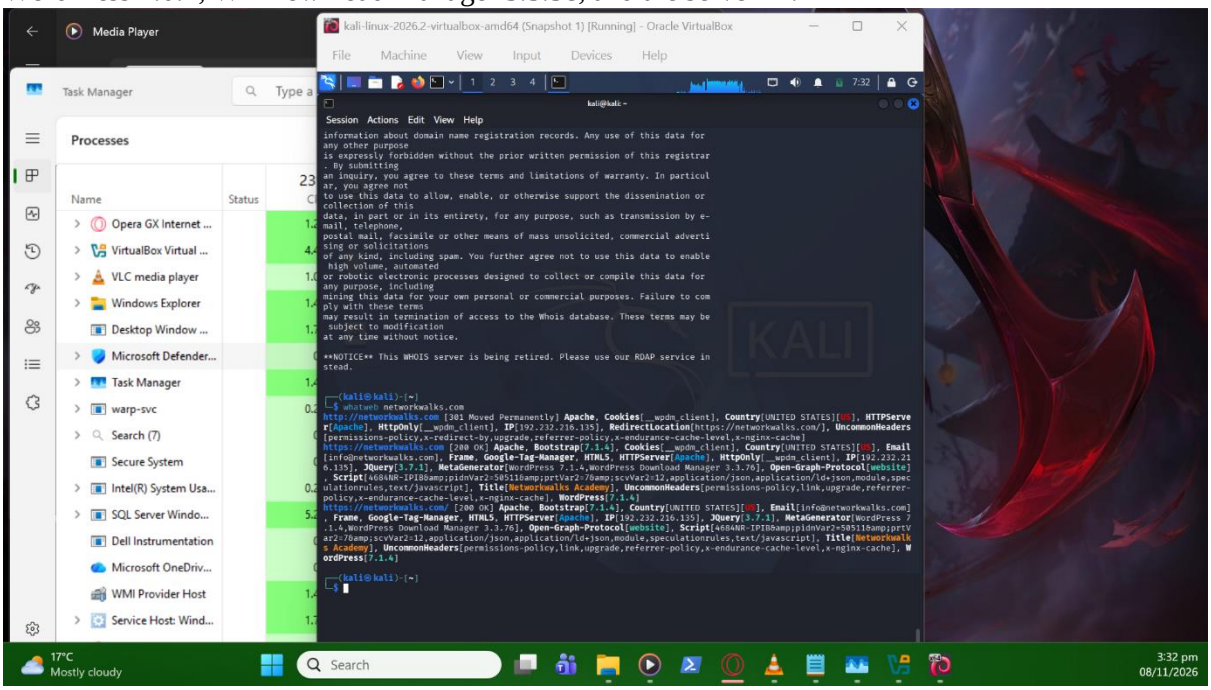
I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools: WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon. Each tool was used to collect a different type of information about the target.

- **WHOIS:** Used to obtain publicly available domain registration information. The results revealed the registrar, creation date, and name servers pointing to HostGator.

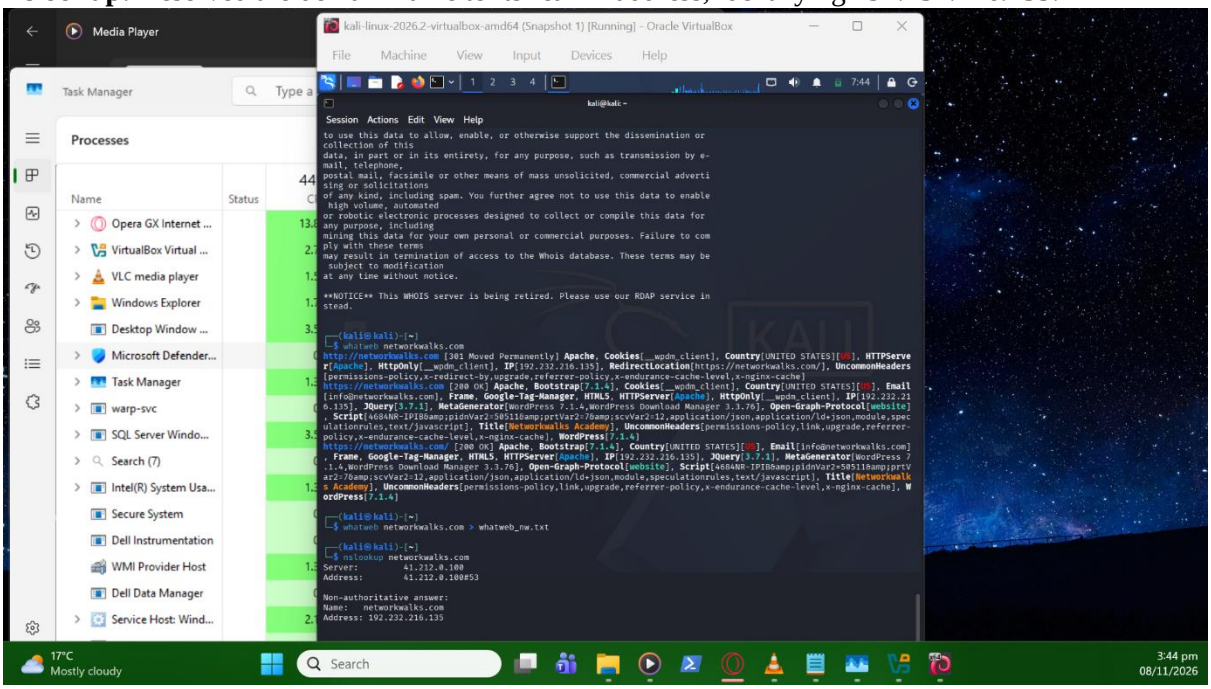


```
The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
Domain Name: networkwalks.com
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: https://www.godaddy.com
Updated Date: 2025-11-12T05:08:41Z
Creation Date: 2019-11-06T17:51:46Z
Registrar Registration Expiration Date: 2027-11-06T17:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: +1.4806242505
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferP
rohibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhi
bited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibi
ted
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhi
bited
Registry Registrant ID: Not Available From Registry
Registrant Name: Registration Private
Registrant Organization: Domains By Proxy, LLC
Registrant Street: DomainsByProxy.com
Registrant Street: 100 S. Mill Ave, Suite 1600
Registrant City: Tempe
Registrant State/Province: Arizona
Registrant Postal Code: 85281
Registrant Country: US
Registrant Phone: +1.4806242599
Registrant Phone Ext:
Registrant Fax:
Registrant Fax Ext:
Registrant Email: https://www.godaddy.com/whois/results.aspx?domain=networkwa
lks.com&action=contactDomainOwner
Registry Tech ID: Not Available From Registry
Tech Name: Registration Private
Tech Organization: Domains By Proxy, LLC
Tech Street: DomainsByProxy.com
Tech Street: 100 S. Mill Ave, Suite 1600
Tech City: Tempe
Tech State/Province: Arizona
Tech Postal Code: 85281
Tech Country: US
Tech Phone: +1.4806242599
Tech Phone Ext:
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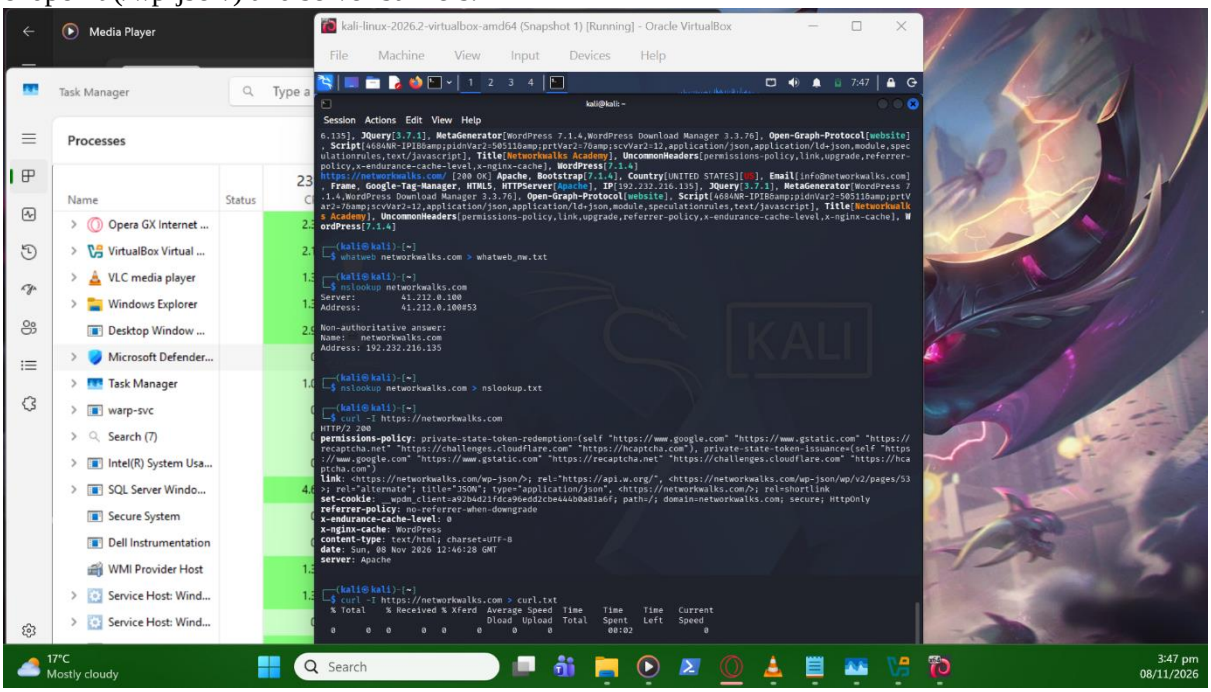
- **WhatWeb:** Used to identify technologies used by the website. The results identified WordPress 7.0.4, WP Download Manager 3.3.58, and the server IP.



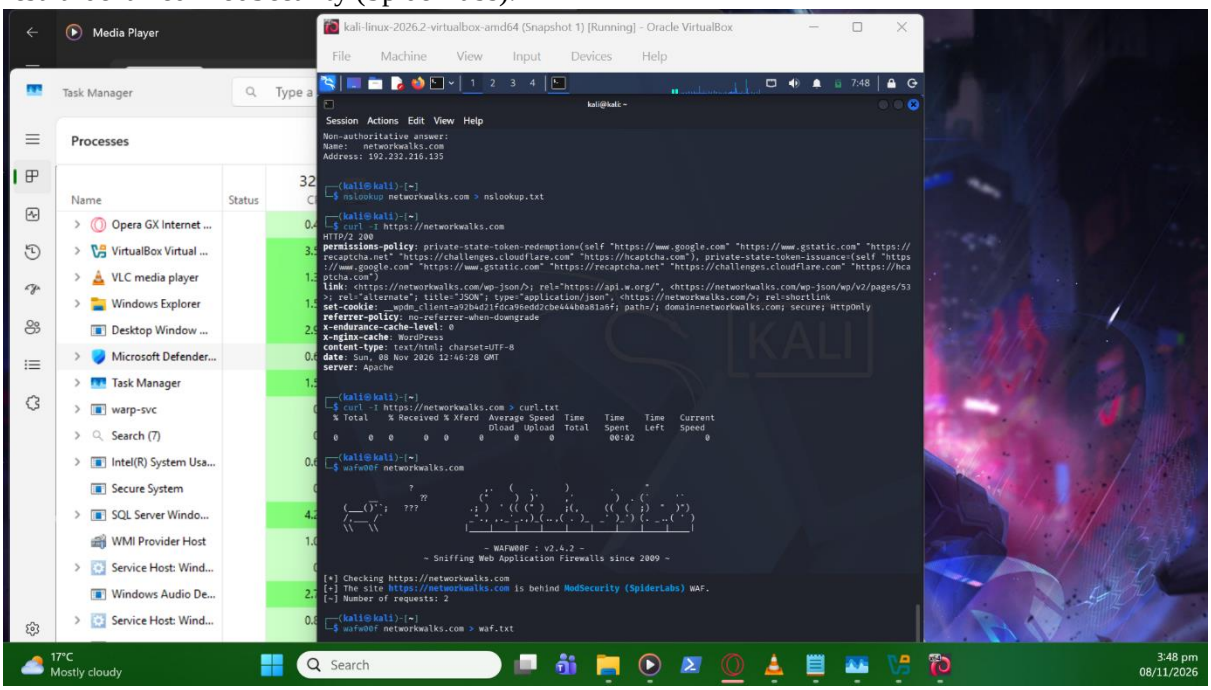

- **Nslookup:** Resolved the domain name to its real IP address, identifying 192.232.216.135.



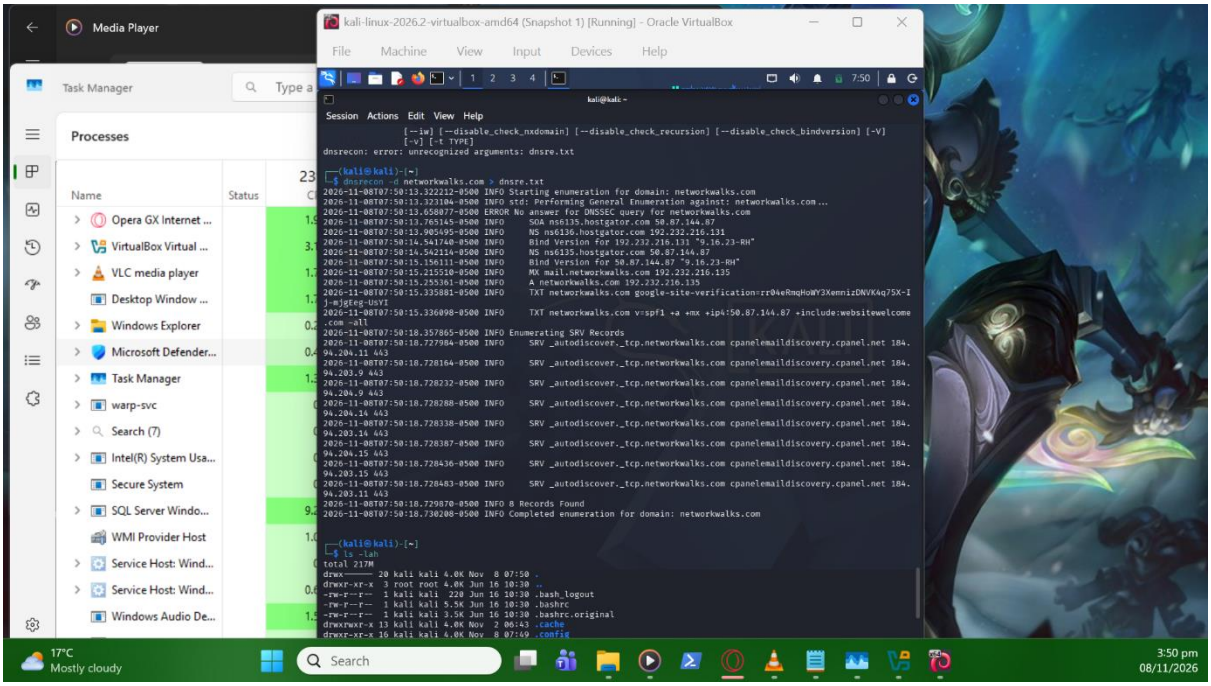
- **Curl -I:** Inspected the HTTP response headers, which exposed the WordPress REST API endpoint (/wp-json/) and server banners.




- **Wafw00f:** Determined whether a Web Application Firewall was protecting the website. The result identified ModSecurity (SpiderLabs).



- **DNSRecon:** Enumerated DNS records, providing information relating to name servers, mail servers, SPF/TXT records, and SRV records.




### 4.2 Network Scanning with Zenmap (W2-PM5)

For the second activity, I used Zenmap to perform active network discovery on my local network. The practical required me to identify my local IP address and subnet, discover live hosts, identify their IP and MAC addresses, and generate a network topology.

1. I first used the Windows ipconfig command to identify my local IP address and LAN subnet.
   [INSERT IPCONFIG SCREENSHOT HERE]

2. I then entered my subnet into Zenmap and selected the **Ping Scan** profile to identify active hosts.
   [INSERT ZENMAP PING SCAN SCREENSHOT HERE]

3. The scan identified several live hosts (including my own PC). I recorded their IP addresses and MAC addresses.

4. After completing the scan, I opened the **Topology** section in Zenmap, enabled the legend, and saved the network topology in PDF format.
   [INSERT ZENMAP TOPOLOGY SCREENSHOT HERE]

**5. Risk Analysis / Impact**

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

| # | Risk / Finding | Evidence / Observation | Potential Impact | Risk Level |
|---|---|---|---|---|
| 1 | Web technology info exposed | WhatWeb identified WordPress and WP Download Manager versions. | Attackers may use exposed version info to identify known exploits. | Medium |
| 2 | Server IP address identifiable | Nslookup resolved the domain to 192.232.216.135. | Provides information about the network location of the web service. | Low |
| 3 | HTTP technical info exposed | Curl returned headers and exposed /wp-json/. | May assist technology fingerprinting and further enumeration. | Low |
| 4 | WAF technology identifiable | Wafw00f identified ModSecurity (SpiderLabs). | Reveals security architecture; attackers know they must bypass it. | Low |
| 5 | DNS infrastructure info exposed | DNSRecon identified DNS, mail and service-related records. | DNS info can help build a broader infrastructure profile. | Medium |
| 6 | Multiple live hosts visible | Zenmap identified multiple live hosts on the local network. | Unknown/unauthorized devices may be present. | Medium |

*Risk level key: Critical / Medium / Low*
*The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities. No exploitation or vulnerability validation was performed.*

## 6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Review publicly exposed technology information:** Organizations should regularly review what information about their web technologies, CMS, and plugins is publicly visible.
2. **Keep software updated:** CMS platforms, plugins, and other web technologies should be regularly updated and reviewed against current security advisories.
3. **Review HTTP headers:** HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.
4. **Review DNS records regularly:** DNS records should be checked periodically to ensure that only required information and services are publicly exposed.
5. **Properly configure and monitor the WAF:** Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.
6. **Perform regular internal network discovery:** Organizations should periodically scan their own networks to identify active devices.
7. **Investigate unknown devices:** Any unexpected device discovered during network scanning should be investigated and verified.

## 7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance, and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS can provide domain information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can provide additional DNS information.

In the network scanning activity, I used Zenmap to identify my local network configuration and discover active hosts, collecting IP and MAC address information and creating a network topology.

The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses. Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope.

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